



NCH · L2 Inventory Efficiency

CO_10 Nawabi Chai HouseLayer 2 of 3

L2 = costing, leakage, wastage, BoM, recipe-driven RM consumption, variance scoring. It **starts from L1** (the POS sales feed already architected for NCH at </ops/visual/nch-l1/>) — every POS line in L1 deducts named RM from L2's inventory pool. It is **independent of L3** (money-out: purchase, expense, salary, bills) — L2 measures in grams/ml/units, not rupees. The two layers cross-reference only at month-close to compute cost-of-sale.

THE INTELLIGENT CIRCLE · L1 + L2 + L3

Three independent layers — different scoring units, different actors, different cadences, different failure modes — each with its own agent. They cross-reference at **three pre-defined closure points**, never continuously. Without all three, the circle doesn't close and leakage hides.

L1 · INFLOW

✓ Architected · live

"What did I sell?"

Actor: counter cashier (Mujib · Kesmat · Nafees · token-issuer)

Unit: ₹ rupees + payment mode (cash / UPI / Razorpay / aggregator)

Cadence: every minute (per POS order)

Failure mode: cash short, UPI mismatch, PIN-less order

Data: pos.order · pos.order.line · cash_events · money_events

[↗ /ops/visual/nch-l1/](#)

L2 · INVENTORY EFFICIENCY

🔧 This page · spec

"What did I consume?"

Actor: kitchen (cook on bhati, batch-prepper, runner)

Unit: grams / ml / Units · NEVER rupees

Cadence: every 3–4 hours (per batch close-out)

Failure mode: over-pouring, batch yield drift, ad-hoc RM bypass, wastage not logged

Data: rm_products · batch_run · rm_consumption_event · wastage_log

[↓ defined below](#)

L3 · OUTFLOW / MONEY OPS

➡ Next · parallel build

"What did I pay for?"

Actor: Zoya (purchase) · Faheem (settlement) · Owner (approvals)

Unit: ₹ rupees + vendor identity + payment mode

Cadence: daily (PO burst) · weekly (bills) · monthly (salary)

Failure mode: PO never closed, bill never matched, advance never adjusted, vendor overpaid

Data: po_lifecycle · vendor_bills · salary_run · expense

[queued · separate spec](#)

THREE CLOSURE POINTS (THE ONLY PLACE LAYERS CROSS)

1. L1 → L2	"100 Irani Chais sold" (L1) → "should have consumed 6 L milk + 4 L decoction" (L2) → did the actual batch yield support it within 5%	catches kitchen leakage per sale · real-time
2. L2 → L3	"Consumed 80 L milk this month" (L2) × "paid Prabhu ₹2,200/L avg" (L3) = milk cost-of-sale ₹1.76 L	catches purchase leakage per month · cost-of-sale
3. L3 → L1	"Paid vendors ₹X this week" (L3) → next batch of RM available → next chai sold (L1)	catches working-capital leakage per cycle · cash flow

POS ITEMS

34

14 chai · 18 snacks · 1 shared · 1 uncat

SHAPES USED

5 / 10

S1, S2, S3, S5, S7

BATCHES

3

Decoction · Milk · Khajoor

MASTER-CODED

26 / 34

≥ 76% with NCH- or HN-RM-XXX

BOM GAPS

3

Black Coffee · Boost · Roll/Fajita

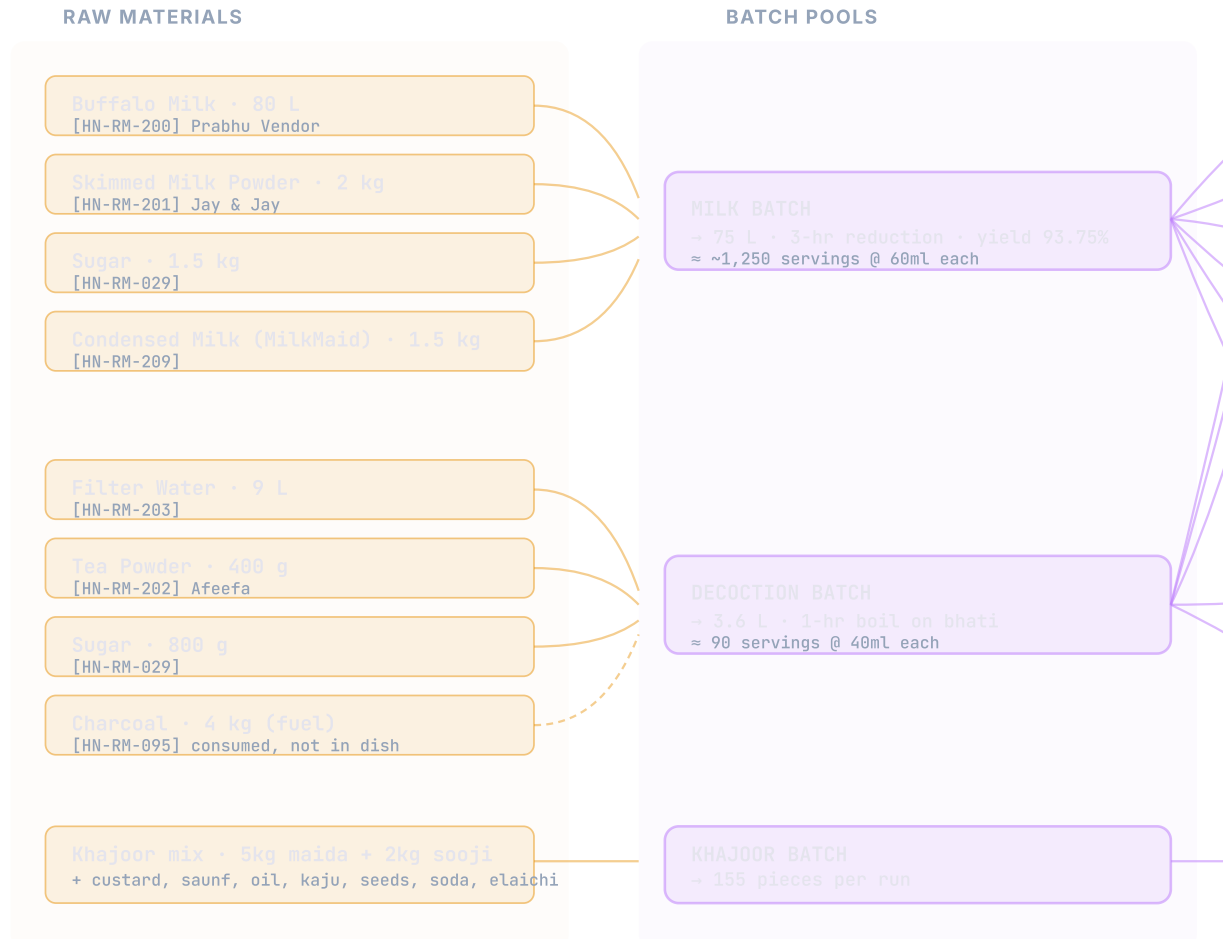
CATALOG DUPS

2+

Code NCH-BM × 2 · 5 dup product IDs

THE TWO-BATCH SPINE — NCH OPERATIONAL LOGIC

NCH is fundamentally a **two-batch business**. The decoction pool and the milk pool are the upstream RM consumers; ~80% of POS items pull a portion from one or both. The remaining ~20% are direct-from-shelf (resale: biscuits, water, niloufer boxes) or simple def+apx wraps (bun + spread). The Khajoor batch is a third, smaller batch that produces 155 pieces per run.



MILK BATCH 75 L YIELD · 93.75%

80 L Buffalo Milk + 2 kg SMP + 1.5 kg Sugar + 1.5 kg Condensed Milk
→ 3-hr reduction → 75 L Milk Batch

DECOCTION BATCH 3.6 L · ~90 SERVINGS

9 L Filter Water + 400 g Tea Powder + 800 g Sugar
→ 1-hr boil on bhati (4 kg charcoal) → 3.6 L Decoction

KHAJOOR BATCH 155 PCS / RUN

5 kg Maida + 2 kg Sooji + 200 g Custard Powder + 100 g Saunf + 6 L Oil
+ 200 g Kaju + 200 g Pumpkin Seeds + 30 g Baking Soda + 50 g Elaichi
→ 155 pieces

SHAPE CATALOG · 10 LOGICAL PERMUTATIONS

Every POS item — current or future — falls into one of these 10 shapes. Adding a new product = pick a shape and fill its parameters. NCH currently uses 5 shapes; the other 5 are future-ready (HE uses all 10).

S1

RESALE

in use

1× DEF

Niloufer biscuits (5 SKUs), Water bottle, Osmania Biscuit, Cheese Balls, Chicken Cutlet, Pyaaz Samosa, Khajor (1pc from batch of 155)

S2

MULTI_UNIT

in use

N× DEF

Osmania Biscuit Pack of 3 — 3 pieces of HN-RM-205

S3

SINGLE_PORTION

in use

1× APX

Irani Milk = 100ml from milk batch + 3g sugar (the milk is the dish)

S4

MULTI_PORTION

unused at NCH

N× APX

(unused at NCH) — HE: Aloo Gobi (70g aloo + 70g gobi)

S5

BATCH+PROTEIN

in use

1× APX-BATCH + 1× APX (or N× APX-BATCH)

Irani Chai = 60ml milk-batch + 40ml decoction-batch · Kadak Chai · Zafrani · Ginger Tea · Chocolate Tea · Lemon Tea · Black Tea · Badam Milk · Horlicks · Boost · Coffee variants

S6

MARINATED_PROTEIN

unused at NCH

1× APX protein + 1× APX-BATCH marinade

(unused at NCH) — HE: Tandoori Chicken, all Tikka, all Kabab

S7 WRAP

in use

1× DEF wrap + 1× APX filling (+ optional APX veg)

Bun Maska = 1 bun + 36g butter · Malai Bun = 1 bun + 100g malai · Nutella Bun = 1 bun + 15g nutella · Bun Maska Jam = 1 bun + 50g jam · Cream Bun (BoM gap)

S8 UNIT_PROTEIN

unused at NCH

1× DEF organ (+ optional batch)

(unused at NCH) — HE: Mutton Brain, Mutton Brain Dry

S9 COMPOSITE

unused at NCH

Σ component POS items

(unused at NCH today) — HE: Chicken Biryani + 2pc Kabab, all CM01–05, Family Combo

S10 NO_RM

unused at NCH

∅ (empty)

(no charge lines on NCH POS today) — HE: Packaging Charges, Delivery Charges

ALL NCH POS ITEMS, GROUPED BY SHAPE

Each item card shows the shape badge, master code (or ad-hoc flag), price, and atomic recipe.

S5 BATCH+PROTEIN		13 items	
Irani Chai id 1028 · NCH-IC · ₹20 60 ml Milk Batch APX 5% BATCH ad-hoc 40 ml Decoction Batch APX 5% BATCH ad-hoc		Kadak Chai id 1632 · NCH-KC · ₹20 40 ml Milk Batch APX 5% BATCH ad-hoc 60 ml Decoction Batch APX 5% BATCH ad-hoc	
Zafrani Chai id 1633 · NCH-ZC · ₹50 60 ml Milk Batch APX 5% BATCH ad-hoc 40 ml Decoction Batch APX 5% BATCH ad-hoc 0.04 g Saffron (Kesar) APX 15% HN-RM-216		Irani Ginger Tea id 1634 · NCH-IGT · ₹20 60 ml Milk Batch APX 5% BATCH ad-hoc 40 ml Decoction Batch APX 5% BATCH ad-hoc 1 g Ginger APX 15% HN-RM-059	
Irani Black Tea id 1635 · NCH-IBT · ₹20 100 ml Decoction Batch APX 5% BATCH ad-hoc 3 g Sugar (extra) APX 10% HN-RM-029		Irani Chocolate Tea id 1636 · NCH-ICT · ₹30 60 ml Milk Batch APX 5% BATCH ad-hoc 30 ml Decoction Batch APX 5% BATCH ad-hoc 10 g Chocolate APX 10% ad-hoc BoM gap	
Lemon Tea id 1085 · ₹20 10 ml Decoction Batch APX 5% BATCH ad-hoc		Irani Badam Milk id 1640 · NCH-IBM · ₹30 87 ml Milk Batch APX 5% BATCH ad-hoc	

5 g

Sabja Seeds (Basil)

APX 10%

HN-RM-217

1 g

Ginger

APX 15%

HN-RM-059

0.25 Units

Lemon (Nimbu)

DEF

HN-RM-065

3 g

Sugar (extra)

APX 10%

HN-RM-029

10 g

Almonds (Badam)

APX 10%

HN-RM-117

Irani Horlicks

id 1637 · NCH-IH · ₹30

S5

87 mL

Milk Batch

APX 5%

BATCH

ad-hoc

3 g

Sugar (extra)

APX 10%

HN-RM-029

10 g

Horlicks

APX 10%

ad-hoc

BoM gap

Irani Boost

id 1638 · NCH-IB · ₹30

S5

87 mL

Milk Batch

APX 5%

BATCH

ad-hoc

3 g

Sugar (extra)

APX 10%

HN-RM-029

10 g

Boost

APX 10%

ad-hoc

BoM gap

Nawabi Special Coffee

id 1084 · NCH-NSC · ₹30

S5

98 mL

Milk Batch

APX 5%

BATCH

ad-hoc

1 Units

Coffee Pouch

DEF

HN-RM-218

Zafrani Coffee

id 1642 · NCH-ZCF · ₹60

S5

98 mL

Milk Batch

APX 5%

BATCH

ad-hoc

1 Units

Coffee Pouch

DEF

HN-RM-218

0.04 g

Saffron (Kesar)

APX 15%

HN-RM-216

Black Coffee

id 1639 · NCH-BC · ₹30

S5

100 mL

Hot Water

APX 5%

ad-hoc

BoM gap

1 Units	Coffee Pouch	DEF	HN-RM-218	BoM gap
3 g	Sugar	APX 10%	HN-RM-029	BoM gap

S7 WRAP

5 items

Bun Maska (Maska Bun) id 1029 · NCH-BM · ₹40	S7	Bun Maska Jam id 1644 · NCH-BMJ · ₹50	S7
1 Units Buns	DEF HN-RM-204	1 Units Buns	DEF HN-RM-204
36 g Amul Butter	APX 10% HN-RM-046	50 g Jam	APX 10% ad-hoc BoM gap
Malai Bun id 1100 · NCH-BM · ₹30	S7	Cream Bun id 1645 · NCH-CRB · ₹50	S7
1 Units Buns	DEF HN-RM-204	1 Units Buns	DEF HN-RM-204
100 g Malai (clotted cream)	APX 10% ad-hoc BoM gap	50 g Cream	APX 10% ad-hoc BoM gap
Nutella Bun id 1643 · NCH-NB · ₹60	S7		
1 Units Buns	DEF HN-RM-204		
15 g Nutella	APX 10% ad-hoc BoM gap		

S1 RESALE

14 items

Khajor id 1435 · NCH-KH · ₹25	S1	Osmania Biscuit id 1030 · NCH-OB · ₹8	S1
1 Units Khajor (1 piece from		1 Units Osmania Biscuit (Loose)	DEF HN-RM-205

1 Units

batch of 155)

DEF

BATCH

ad-hoc

Niloufer Osmania 500g

id 1093 · NCH-OBBOX · ₹250

S1

1 Units

Osmania Biscuit Box (Niloufer)

DEF

HN-RM-208

Niloufer DCC 75g

id 1383 · HN-RM-221 · ₹120

S1

1 Units

Niloufer DCC 75g

DEF

HN-RM-221

Niloufer Fruit 100g

id 1384 · HN-RM-222 · ₹70

S1

1 Units

Niloufer Fruit 100g

DEF

HN-RM-222

Niloufer Fruit 200g

id 1385 · HN-RM-223 · ₹150

S1

1 Units

Niloufer Fruit 200g

DEF

HN-RM-223

Niloufer Osmania 100g

id 1405 · ₹60

S1

1 Units

Niloufer Osmania 100g

DEF

ad-hoc

BoM gap

Chicken Cutlet

id 1031 · NCH-CC · ₹25

S1

1 Units

Chicken Cutlet (Unfried)

DEF

HN-RM-206

Cheese Balls

id 1099 · NCH-CB · ₹50

S1

1 Units

Cheese Balls Raw

DEF

HN-RM-219

Pyaaaz Samosa

id 1097 · NCH-PS · ₹15

S1

1 Units

Samosa Raw

DEF

HN-RM-210

Chicken Roll (with mayo)

id 1374 · NCH-CMR · ₹50

S1

1 Units

Chicken Roll (pre-made or assembled?)

DEF

ad-hoc

BoM gap

Chicken Fajita

id 1376 · NCH-FAJ · ₹30

S1

1 Units

Chicken Fajita (pre-made?)

DEF

ad-hoc

BoM gap

Water

id 1076 · NCH-WTR · ₹10

S1

1 Units

Bottled Water
500ml

DEF

HN-RM-207

Osmania Biscuit Box (Niloufer)
(uncategorised)

id 1092 · HN-RM-208 · ₹0

S1

1 Units

Osmania Biscuit
Box (Niloufer)

DEF

HN-RM-208

S2

MULTI_UNIT

1 items

Osmania Biscuit · Pack of 3

id 1033 · NCH-OB3 · ₹20

S2

3 Units

Osmania Biscuit
(Loose)

DEF

HN-RM-205

S3

SINGLE_PORTION

1 items

Irani Milk

id 1641 · NCH-IM · ₹20

S3

100 ml

Milk
Batch

APX 5%

BATCH

ad-hoc

3 g

Sugar
(extra)

APX 10%

HN-RM-029

CLEANUP CHECKLIST BEFORE AUTOMATION LOCKS DOWN

3 BOM GAPS · NEED STAFF INPUT (MUJIB / TANVEER)

- **Black Coffee** [1639] — qty of coffee pouch + sugar not in PDF.
- **Irani Boost** [1638] — recipe inferred (Horlicks-pattern: 87ml milk + 3g sugar + 10g Boost). Confirm.
- **Chicken Roll / Chicken Fajita / Cream Bun** [1374, 1376, 1645] — composition unclear (resale vs in-house assembly + qty of filling).

CATALOG DEDUP · BEFORE LOCKING MASTER

- **Code NCH-BM** shared by Bun Maska (1029) AND Malai Bun (1100). Will collide on barcode lookup.
- Recent NCH POS sales reference duplicate IDs 1102 / 1118 / 1115 / 1453 / 1655 for items already present at lower IDs. Dedup before custom-layer cutover.
- **Haleem family (5 SKUs)** exists in Odoo with `available_in_pos = false`. Currently sold via WhatsApp/manual flow → invisible to POS variance logic. Either POS-enable Haleem or build a parallel non-POS sales feeder.

L2 STACK · CUSTOM BUILD TARGET (INDEPENDENT OF L3)

The L2 layer — RM master, recipes (10 shapes), batch runs, per-sale RM consumption, variance scoring — moves to **D1 + Cloudflare Functions**. L3 (vendor bills, salary, expense) lives in its own parallel D1 schema; the two cross-reference only at month-close on the **L2 → L3 closure point**. Odoo (odoo.hnhotels.in) retreats from operational truth to nightly compliance posting. POS Odoos (test/ops.hamzahotel.com) stay untouched — they remain L1's sales-truth source.

D1 SCHEMA SKETCH · 6 TABLES DRIVE THE ENTIRE LAYER

```

-- 1. Canonical RM master (replaces Odoo product.template + product.category)
CREATE TABLE rm_products (
  id          INTEGER PRIMARY KEY,
  code        TEXT UNIQUE,      -- HN-RM-XXX or NCH-XX
  name        TEXT,
  category    TEXT,
  uom         TEXT,             -- kg, g, L, ml, Units, can ...
  current_cost_paise INTEGER,    -- live vendor price
  vendor_id   INTEGER,
  is_batch_intermediate BOOLEAN -- true for "Milk Batch", "Decoction", "Khajoo
);

-- 2. Per-RM tolerance bands (variance scoring)
CREATE TABLE rm_tolerance (
  rm_id        INTEGER REFERENCES rm_products(id),
  tolerance_pct INTEGER          -- 0 for DEF, 5-15 for APX
);

-- 3. POS recipe = which shape + parameters
CREATE TABLE pos_recipe (
  pos_item_id  INTEGER PRIMARY KEY, -- maps to product_id from POS Odoo
  brand        TEXT,                 -- 'NCH' or 'HE'
  shape_code   TEXT,                 -- 'S1'..'S10'
  params_json  TEXT                  -- shape-specific component list
);

-- 4. Production batch run (NEW concept — doesn't exist in Odoo today)
CREATE TABLE batch_run (
  id          INTEGER PRIMARY KEY,
  batch_rm_id INTEGER REFERENCES rm_products(id),
  brand       TEXT,
  started_at  TEXT,
  completed_at TEXT,
  expected_yield REAL,              -- 75 L for Milk, 3.6 L for Decoction
  actual_yield REAL,                -- staff-logged at end of cook
  prepared_by_pin TEXT
);

-- 5. Inputs to a batch run (denorm of recipe at run time)
CREATE TABLE batch_run_input (
  batch_run_id INTEGER REFERENCES batch_run(id),
  rm_id        INTEGER REFERENCES rm_products(id),

```

```

    qty_used      REAL,
    uom           TEXT
);

-- 6. Per-sale RM consumption (one row per recipe component per sale)
CREATE TABLE rm_consumption_event (
    id             INTEGER PRIMARY KEY,
    pos_order_line_id TEXT,          -- from POS Odoo
    occurred_at    TEXT,
    rm_id          INTEGER,
    source_batch_run_id INTEGER,     -- NULL if direct-RM (no batch)
    expected_qty   REAL,             -- from recipe
    actual_qty     REAL,             -- system-deducted = expected (no live measure
    variance_qty   REAL              -- computed at batch close-out
);

```

DAILY FLOW · HOW VARIANCE GETS CAUGHT AT THE MOMENT OF ERROR

- 1. Batch start (e.g., 6:30am):** Mujib opens "Start Decoction Batch" on PIN-gated UI. System creates batch_run with expected_yield = 3.6 L, decrements rm_products on-hand for 9 L water + 400g tea + 800g sugar (the inputs).
- 2. Batch end (~7:30am):** Mujib enters actual yield (e.g., "3.4 L"). System computes batch yield variance = $(3.4 - 3.6) / 3.6 = -5.5\%$ — within tolerance. Logged to batch_run.actual_yield.
- 3. Throughout the day:** Each Irani Chai sale at the POS triggers rm_consumption_event rows: 60ml from Milk Batch + 40ml from Decoction Batch. The batch's "remaining" auto-decrements.
- 4. Batch close-out (when remaining = 0):** System checks: did the actual servings × per-cup ml match the actual yield? If servings × 40ml > actual yield + tolerance, that's **over-dispensing** (free chai → leakage). If < actual yield – tolerance, **under-dispensing** or unrecorded sale (cash in pocket).
- 5. Variance alert:** Agent fires WABA to Mujib first ("Decoction batch closed with -8% variance: 80 servings sold but yield 3.4L = 85 servings expected"). 30 min SLA → escalate to owner.

WHY L2 LEAVES ODOO (L2-SPECIFIC REASONS)

- **No mrp.bom on this Odoo** — manufacturing module not installed → can't model the two-layer BoM (Decoction batch + per-cup portion) at all.
- **No first-class concept of "batch run"** — Odoo has no native "Mujib started a 9L decoction batch at 6:30am, finished at 7:30am with 3.4L actual yield" event. We'd have to misuse mrp.production or invent custom models — both fragile.
- **Variance & WABA-alert agents need millisecond-grade reads** from a local D1 — not JSON-RPC round-trips to a hosted Odoo. Per-sale recipe deduction at POS scale needs < 50ms; Odoo can't.
- **10-shape recipe vocabulary** doesn't fit Odoo's BoM model cleanly. Custom pos_recipe table with shape_code + params_json is dead simple in D1.

- **Cost-of-sale crosses into L3** at month-close, not continuously. Keeping L2 in D1 means the L1↔L2 closure is fast and L2↔L3 is a once-a-month batch — no live coupling needed.
- **POS Odoos (HE prod, NCH prod) remain untouched** as L1's source. Compliance Odoo (odoo.hnhotels.in) keeps GST/audit ledger only.

Generated from ops.hamzahotel.com POS configs (NCH = company_id 10) + Master Production Sheet · 5 May 2026

34 active POS items · 3 batch constants · 5 of 10 shapes used · 3 BoM gaps · 2+ catalog dedups